

BIC: GenAI Accelerator Program

LLM Landscape and Model Selection

21st September 2026

Hi, I am Tanmay

- 2x Founder - AI Consultant and Trainer
- Partnered with global enterprises, financial institutions, VC-backed startups, educational institutions, and leading nonprofits
- Ex-Microsoft: Built search and recommendation systems for Bing & Microsoft Teams
- B.Tech and M.Tech (Dual Degree) from IIT Madras in Computer Science and Engineering

Microsoft ebay POLYMATH

LLM Landscape

Understanding Model options, Selecting the correct model, Tracking model progress

Question

What are some LLMs that you have heard of?

Closed Source (Proprietary) vs Open Source Models

Open-Source LLMs	Proprietary LLMs
Model weights/parameters are publicly available	Model weights/ parameters are private
Can be downloaded, fine-tuned, and deployed locally.	Access only via an API or hosted service.
Greater flexibility & customization, but require more infrastructure and expertise to run.	Easier to use and integrate, but limited transparency, control, and adaptability.














Examples - Closed Source vs Open Source Models

Open-Source LLMs	Proprietary LLMs
Meta: LLaMA-2, LLaMA-3, LLaMA-4	**OpenAI**: GPT-4.1, GPT-4o, GPT-5, o3, o4-mini
OpenAI: gpt-oss-120B, gpt-oss-20B	**Anthropic**: Claude 4 (Opus/Sonnet), Claude 3.7 (Sonnet)
DeepSeek: DeepSeek-V3, DeepSeek-R1	**xAI**: Grok 3, Grok 4 family
Alibaba: Qwen3:8b, Qwen 3-235B-A22B	**Google DeepMind**: Gemini 3, Gemini 2.5 Pro, Gemini 2.5 Flash-Lite/Flash
Mistral: Mistral Large 2, Mixtral 8x7B, Mixtral 8x22B	**Amazon**: Nova series (Nova Micro, Nova Lite, Nova Pro)

Question

Why is the term open-source LLM misleading?

- Only the model weights are released
- Training code, data and pipeline stay private
- No way to contribute back, unlike real open source

Hence, *open-weights* is a more suitable term.

Selecting the right LLM

Choosing the right model

- **Evaluate Task Fit:** Choose an LLM based on its proven strengths in tasks similar to those required for your project, such as text generation, summarisation, question-answering, tool calling.

- **Evaluate Data Fit:** Confirm that the LLM is well-suited for your data characteristics (e.g., language, domain specificity - coding, law, medical, finance etc)

- **Resource considerations**
  - Computing power (GPUs)
  - Cost of training and inferencing

Factors to consider

Factor	What It Means / Why It Matters
Model Size	Larger = higher quality, better reasoning; Smaller = faster, cheaper, good for simple tasks
Context Window	Determines how much information the model can “see” at once (important for RAG & long docs)
Latency & Throughput	Impacts real-time use cases (support, coding copilots, voice) vs. batch workloads
Cost per 1M Tokens	How expensive the model becomes as usage grows
Modalities	Text-only vs. text+image vs. multimodal reasoning
Fine-Tuning Support	Needed for domain adaptation, compliance-heavy or specialized tasks
Safety & Reliability	Hallucination rate, guardrail support, consistency
Open-Source vs Closed-Source	Open-source = control, customization, private hosting; Closed-source = higher performance and lower ops overhead

Open-Source vs Proprietary Models

Decision Factor	Open-Source Models	Closed-Source Models
Control & Customisation	Full control; can fine-tune and modify	Limited customisation
Security & Compliance	Data stays within your environment	Data may leave environment; vendor-controlled
Cost Structure	Lower long-term cost; higher infra & ops	Higher per-token cost; no infra overhead
Performance	Variable; can be domain-tuned	Strong out-of-the-box performance
Reliability & Updates	You manage updates, safety, and versioning	Vendor handles reliability, updates, and safety
Operational Overhead	High – requires ML + DevOps	Low – minimal setup and maintenance
Best Use Cases	Compliance-heavy, domain-specific workloads	General usage, rapid development

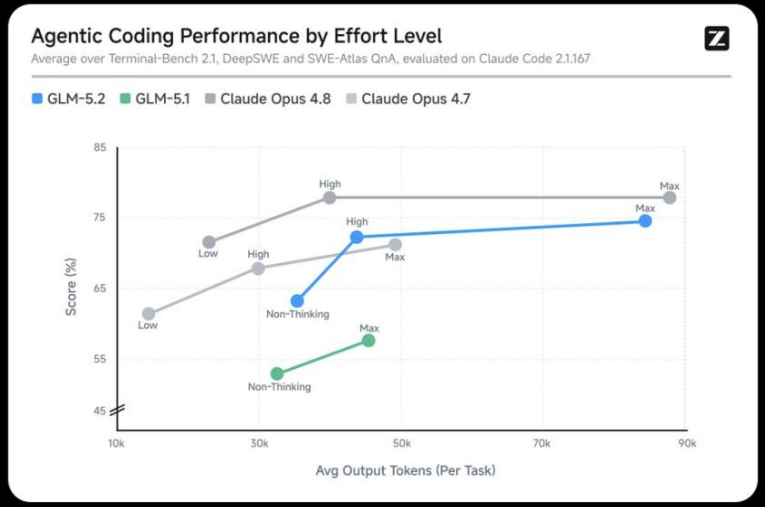
Open-source has almost caught up (1)

Z.ai @Zai_org · Jun 16

Introducing **GLM**-5.2: Frontier Intelligence, Open Weights

- Significant improvements in coding and agentic tasks
- Strong long-horizon capabilities with a 1M context window
- Two levels of reasoning effort: **GLM**-5.2 (max) pushes the limits, while **GLM**-5.2 (high) strikes a strong

Show more



684 2.6K 12K 7.2M

Alex Finn @AlexFinn Subscribe

I was wrong

I've been saying for months that open source AI models are 6 months behind frontier

They caught up. GLM 5.2 is as good as Opus 4.8

This changes everything. If you run GLM 5.2 locally no government can take it away. You become sovereign

And even if you run through APIs, its a fraction of the cost

Marc Andreessen @pmarca · 22h

Many smart people/AI insiders are saying **GLM**-5.2 is the first Chinese AI model to match and often beat the American big lab public AI models with no compromises. Incredible timing given current events.

401 687 8.3K 704K

Gergely Orosz @GergelyOrosz · Jun 27

This is from a popular inference provider

GLM-5.2 plus the US banning the most capable new models means open source caught up to SOTA closed source coding models

This could be v problematic for Anthropic's and OpenAI's revenue projections and IPO plans...

> **Madison Kanna** @Madisonkanna · Jun 27
> Such a vibe shift happening right now. We've gotten nonstop requests from people who want to start using GLM-5.2. Incredible few weeks for open source models.

49 90 1.5K 175K

Open-source has almost caught up

Arena.ai ✓
@arena

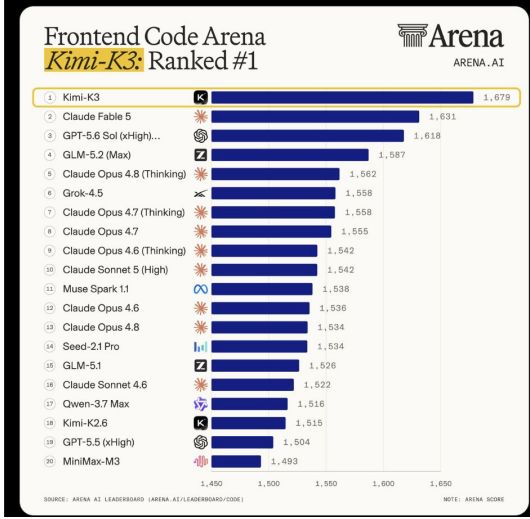
Big news: Kimi-K3 by @Kimi_Moonshot is now #1 in the Frontend Code Arena with 1679 pts, surpassing Claude Fable 5.

This is a 17-place jump from Kimi-k2.6 (#18 -> #1).

In Frontend, Kimi-K3 ranked #1 in 6 of 7 domains: Brand & Marketing, Reference-Based Design, Data & Analytics, Consumer Product, Simulations, and Content Creation Tools, landing #2 only in Gaming behind Fable 5.

The full model weights will be released by July 27.

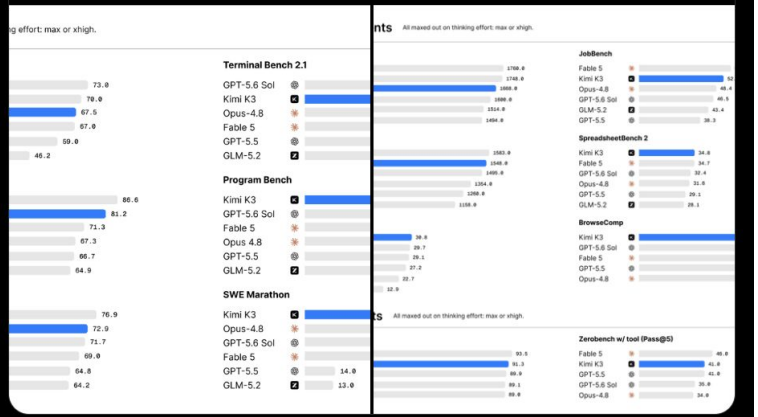
Congrats to the @Kimi_Moonshot team on this major milestone!



Xiaoyin Qu ✓
@quxiaoyin

Kimi just surpasses Fable across multiple benchmarks, and it's open weight! China has caught up.

> **Kimi.ai** ✓ @Kimi_Moonshot · Jul 17
> Introducing Kimi K3: Open Frontier Intelligence
>
> ◆ 2.8 Trillion Parameters, 1 Million Context, Native Multimodal
> ◆ Kimi Delta Attention enables up to 6.3x faster decoding in million-token contexts...



1:00 AM · Jul 17, 2026 · **79K** Views

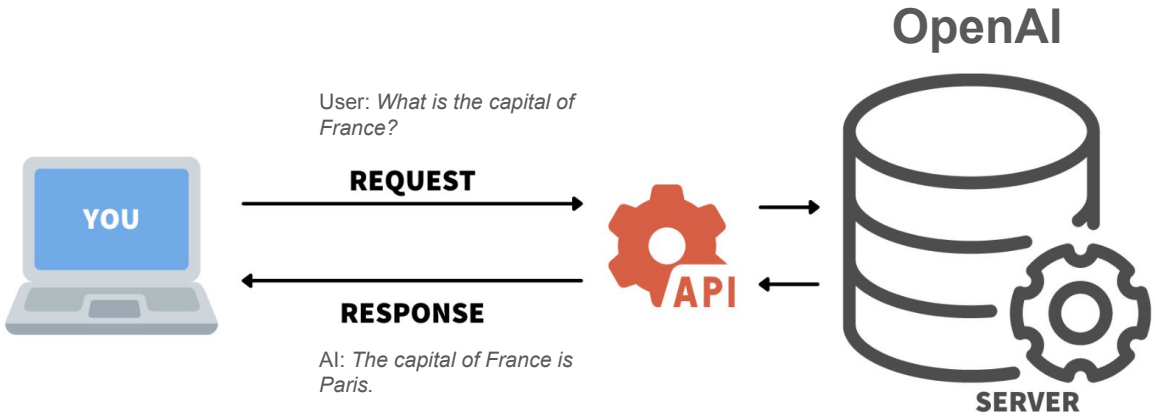
Proprietary models are accessed through APIs

Hosted by the provider - Models like GPT-4 run on OpenAI's infrastructure.

Access via API - You send data (text, image, etc.) over the internet, get results back.

Ready to use - Start building without worrying about infrastructure, hosting or training.

Proprietary models offer ease of use and scalability, but you're dependent entirely on the provider.



Open Source Models can be self hosted

You can actually download the model weights. E.g., Llama - Meta | Hugging Face

And then run it either on

1) **Physical Infrastructure (GPUs)** - Purchase and set up your own hardware for full control. (*Unlikely)*
2) **Virtual GPUs** - GPU Cloud services for scalability and flexibility (AWS/Azure/RunPod/ Qubrid/ Modal)
3) **Managed hosting providers** - Hassle-free, fully managed hosting solutions(Fireworks AI/ Together AI/ Replicate)
4) **Your own laptop!** - Suitable for smaller models with minimal requirements

Option 1 - Physical Infrastructure

You build the GPU farm yourself.

- Procuring the chips — GPUs sit in allocation queues, with lead times often measured in quarters rather than weeks.
- Power — a single rack draws more electricity than a small office building, so you are negotiating megawatts with a utility.
- Cooling — at that density air cooling stops working, which means designing and maintaining liquid cooling loops.
- Space and networking — you need the datacentre floor, the racks, and the high-speed fabric to connect it all.
- People — a team has to keep the whole thing running around the clock.

Battle for the GPUs : A scarce resource in 2026

Elon Musk @elonmusk — Subscribe

This weekend, the @xAI team brought our Colossus 100k H100 training cluster online. From start to finish, it was done in 122 days.

Colossus is the most powerful AI training system in the world. Moreover, it will double in size to 200k (50k H200s) in a few months.

Excellent work by the team, Nvidia and our many partners/suppliers.

10:23 PM · Sep 2, 2024 · **17.1M** Views

4.5K 8.4K 73K 3.8K

Sam Altman @sama

we will cross well over 1 million GPUs brought online by the end of this year!

very proud of the team but now they better get to work figuring out how to 100x that lol

3:44 AM · Jul 21, 2025 · **2.8M** Views

1.5K 1.3K 15K 1.1K

REPORT ENTERTAINMENT GAMING

GPU prices are out of control again



An Nvidia RTX 5090 graphics card. Photo by Tom Warren/The Verge

/ Nvidia and AMD GPUs: so much for MSRP!

by Sean Hollister
Apr 26, 2025, 6:46 AM GMT+5:30

48 Comments (48 New)

Sean Hollister is a senior editor and founding member of The Verge who covers gadgets, games, and toys. He spent 15 years editing the likes of CNET, Gizmodo, and Engadget.

Every so often, Central Computers — one of the last remaining dedicated Silicon Valley computer stores — lets subscribers know it's managed to obtain a small shipment of AMD graphics cards. Today, it informed me that I could now purchase a $600 Radeon RX 9070 XT for $850 — a $250 markup.

It's not alone. I just checked every major US retailer and street prices on eBay, and I regret to inform you: the great GPU shortage has returned. Many AMD cards are being marked up $100, $200, $250, even $280. The street price of an Nvidia RTX 5080 is now over $1,500, a full $500 higher than MSRP. And an RTX 5090, the most powerful consumer GPU? You can't even get the $2,000 card for $3,000 today.

MOST POPULAR

1 Apple made a 24k gold and glass

Source: https://www.theverge.com/report/656783/gpu-prices-are-out-of-control-again

NVIDIA - Selling shovels in a gold rush

Stock grew over 70% in the last year



and, over 1085% in the last 5 years



Option 2 - Virtual GPUs

General Purpose Clouds







Specialized AI GPU Clouds







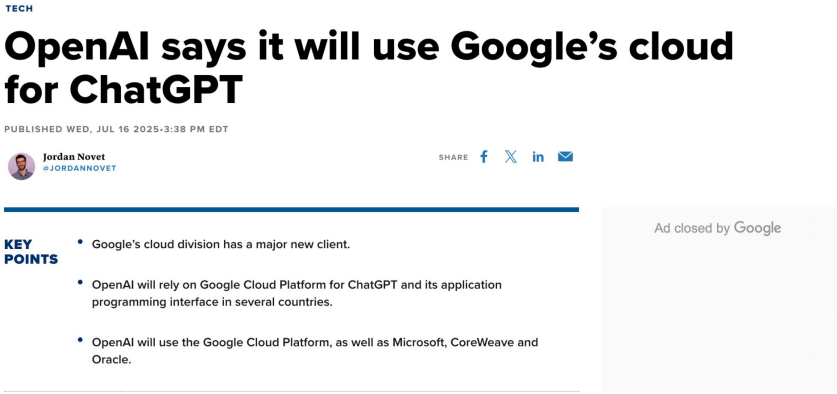


Benefits of Virtual GPUs

- No upfront hardware cost
- No hardware maintenance
- Rapid experimentation
- Global availability
- Scale up instantly
- Pay for what you use

Even the Largest AI Players Rely on Cloud GPUs

Running advanced AI systems requires massive GPU capacity.
Even the biggest AI companies use multiple cloud providers, along with their own GPUs to meet demand, scale quickly, and avoid dependency on a single vendor.



TECH

OpenAI says it will use Google's cloud for ChatGPT

PUBLISHED WED, JUL 16 2025•3:38 PM EDT

Jordan Novet
@JORDANNOVET

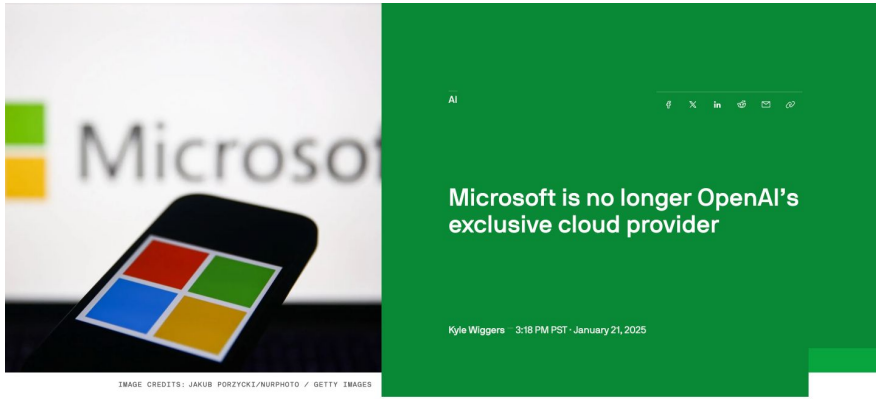
SHARE

KEY POINTS

- Google's cloud division has a major new client.
- OpenAI will rely on Google Cloud Platform for ChatGPT and its application programming interface in several countries.
- OpenAI will use the Google Cloud Platform, as well as Microsoft, CoreWeave and Oracle.

Ad closed by Google

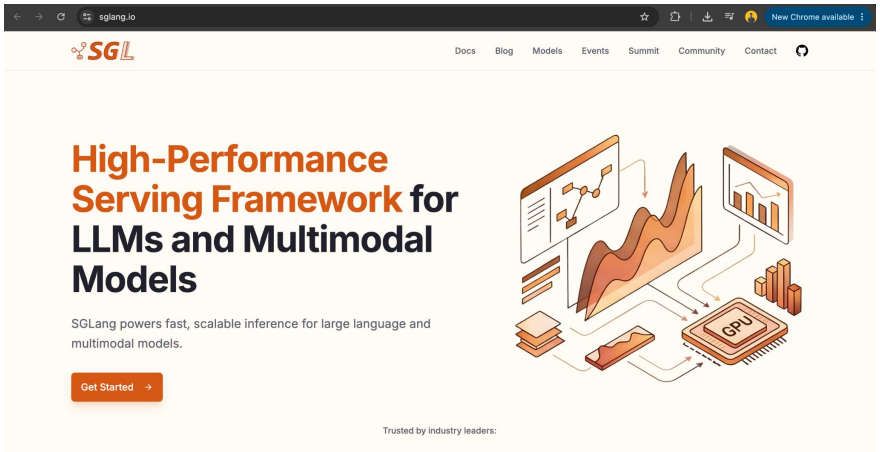
Source:
https://www.cnbc.com/2025/07/16/openai-googles-cloud-chatgpt.html



Source:
https://techcrunch.com/2025/01/21/microsoft-is-no-longer-openais-exclusive-cloud-provider/

Running open source models in production

SGLang



vLLM

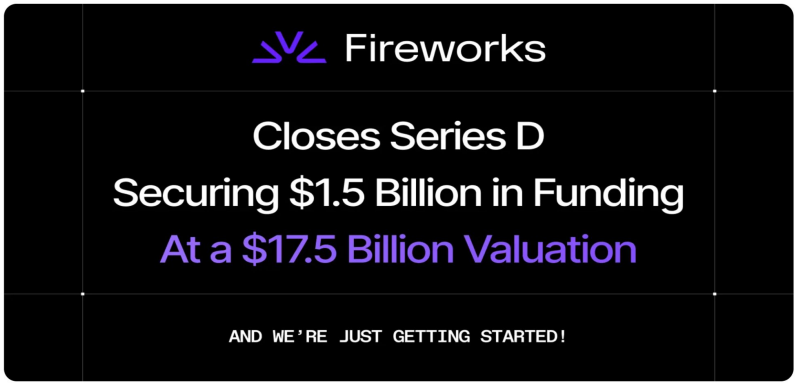


SGLang and vLLM are purpose-built serving frameworks optimised for high throughput, memory efficiency, and concurrent requests.

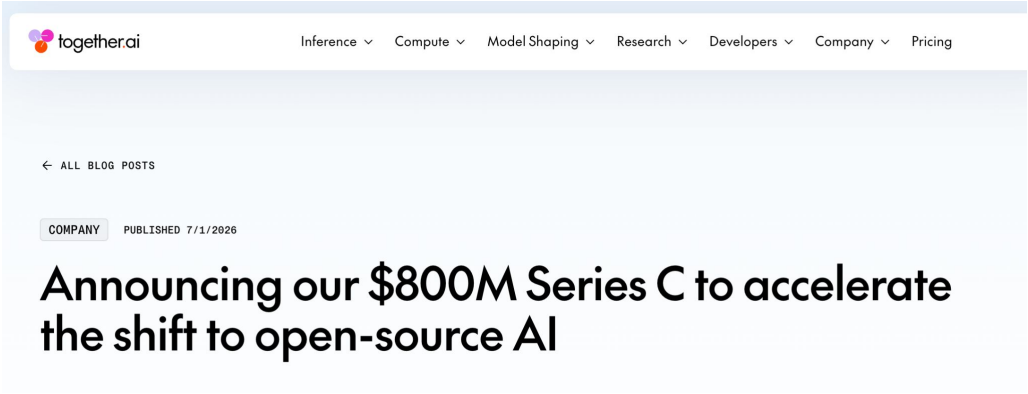
Inference Engineering - getting the most out of your hardware

You can deploy a model as-is and it will work, in production you want more optimization.

Inference engineering is everything you do at deployment to make the same model faster and cheaper on the same hardware.



Every company must own its intelligence

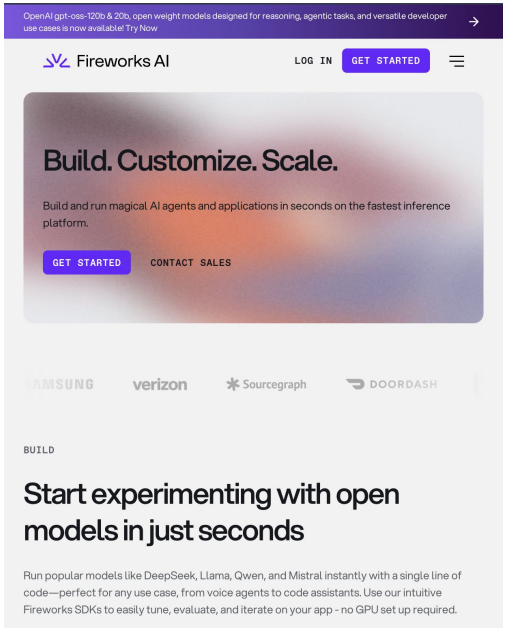


Option 3 - Managed Hosting Providers offer a platform for running LLMs

Managed hosting providers give you the **flexibility of open-source models with the convenience of API-based access.**

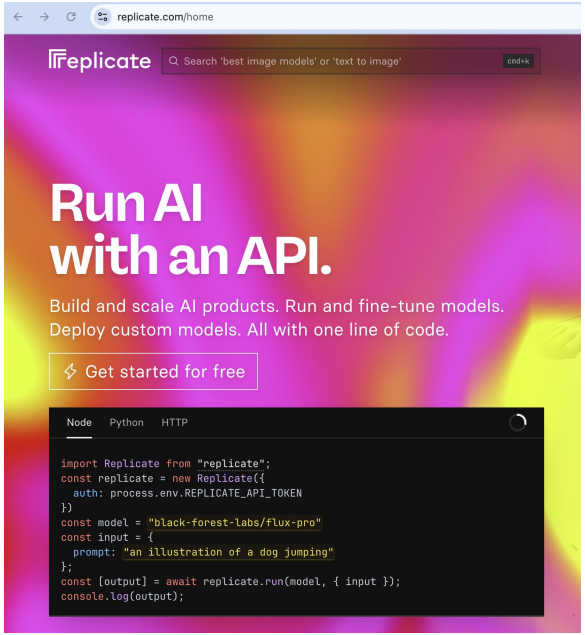
They handle the infrastructure, so you can focus on building and integrating AI into your products.

Fireworks AI



Ref: https://fireworks.ai/

Replicate



Ref: https://replicate.com/

Option 4 - Running LLMs locally

There are now easy tools that can run LLMs on your laptop without the need to set up complicate environments.

This is useful for quick prototyping and experimenting with ideas.

For privacy-conscious users, they can run entirely offline - just switch off your internet and keep working locally.

Ollama



OpenAI and Ollama partner to launch gpt-oss

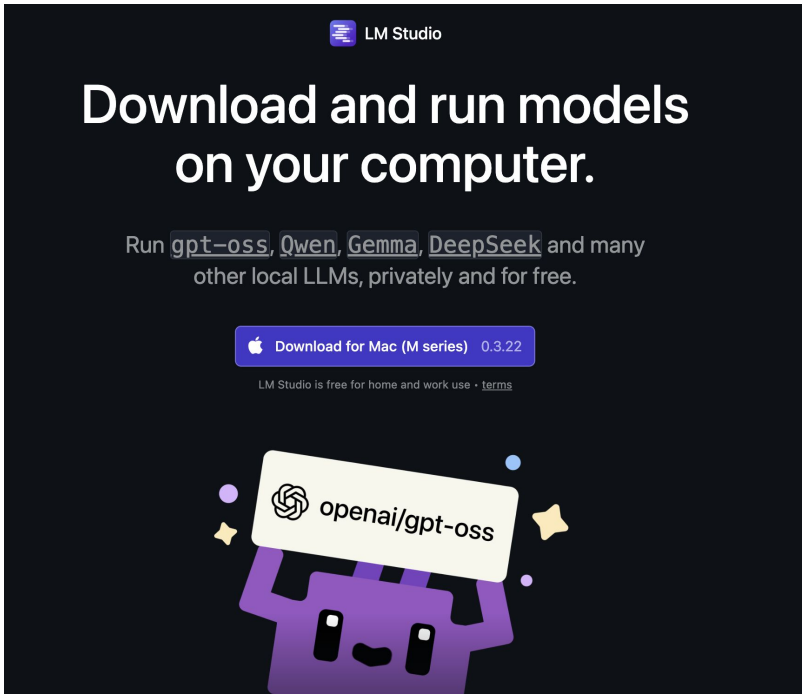
Chat & build with open models

Download

Available for macOS, Windows, and Linux

Ref: https://ollama.com/

LM Studio



Ref: https://lmstudio.ai/

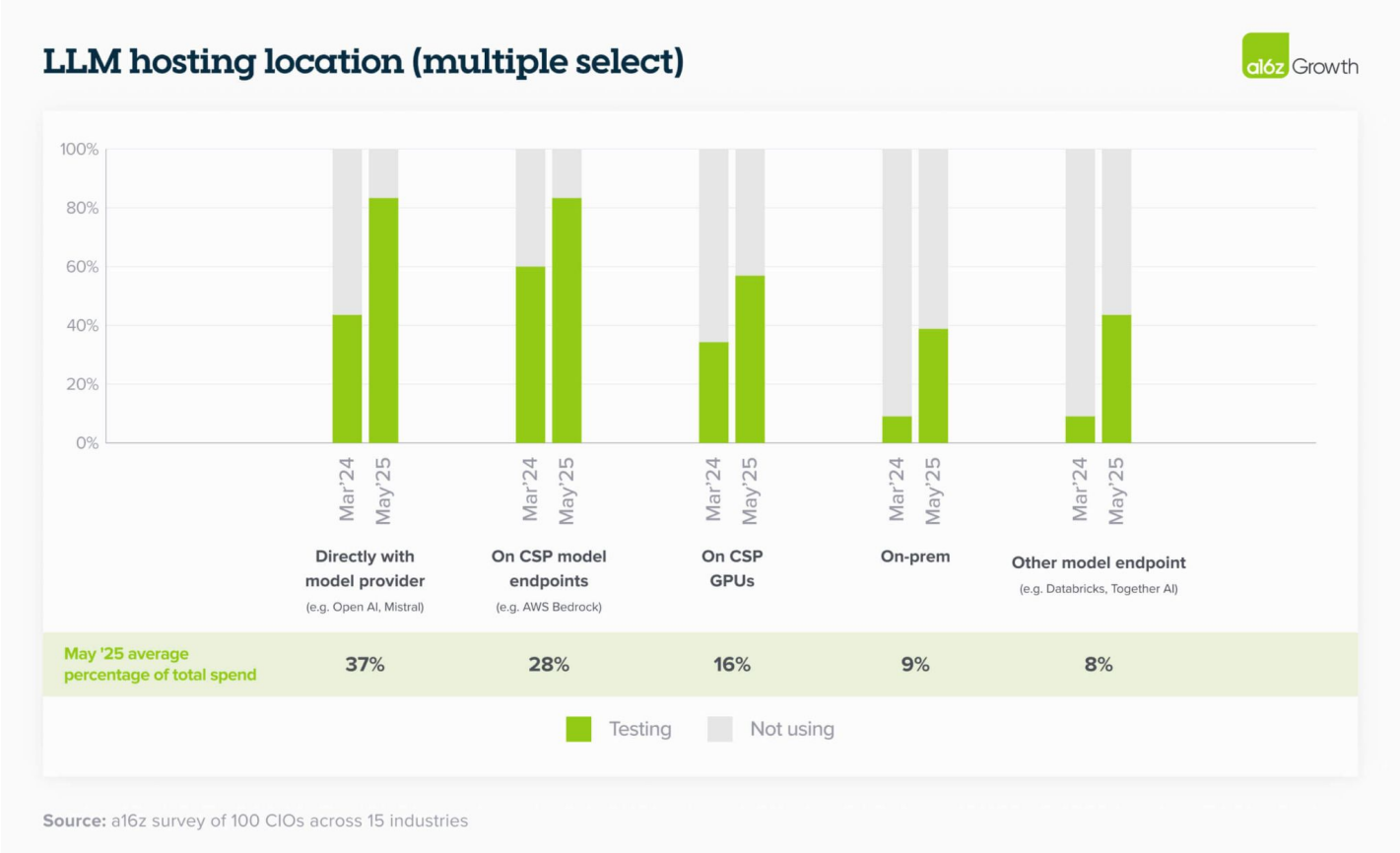
APIs vs Self-Hosting LLMs

APIs	Self-Hosting
No infra setup — provider handles compute, reliability, scaling	You own the infra setup
Costing: per token (input + output)	**Costing:** resource-based (GPU hours consumed)
Proprietary models can only be accessed through APIs, updates happen automatically	Open-Source models can be self-hosted, no auto-updates
Data leaves your environment; SLAs typically bar training on it	Data never leaves your environment — higher compliance
Minimal setup costs, higher running cost	High setup costs - developer costs, infra provisioning etc, typically cheaper at a token level (if optimal utilization)

Comparing different model hosting options

Hosting Option	Cost	Setup Complexity	Scalability	Control & Customization	Typical Use Case
Physical Infrastructure (GPUs)	💰💰💰 High (GPU purchase, maintenance, electricity)	High – need to procure, set up, and maintain hardware	Limited to physical capacity	Full control over hardware, data, and configurations	Large enterprises or research labs with strict data/privacy requirements
Managed Hosting Providers(Fireworks AI, Together AI, Replicate)	💰💰 Medium (pay-per-use or subscription)	Low – provider handles setup, scaling, and maintenance	High – elastic scaling handled by provider	Limited control over backend but can configure models	Startups or teams wanting quick deployment without infrastructure headaches
Virtual GPUs(AWS, Azure, RunPod, Qubrid, Modal)	💰 Medium to High (depends on usage hours)	Medium – need to configure cloud environment	High – can scale up/down GPU capacity	More control than managed hosting, but less than owning hardware	Teams needing bursts of compute or flexible scaling
Your Own Laptop	💰 Low (hardware already owned)	Very Low – local install	Very Limited – depends on laptop specs	Full control locally but very limited model size	Prototyping, experimenting with small models, learning/training demos

Hosting preferences still vary widely, though enterprises have quickly built trust for model providers over the last year



Source
OSS

Choose your LLM strategy

- model class and size / open or proprietary / deployment /

Scenario A: Product Listing Enrichment

The problem: Supplier data arrives with messy titles, missing attributes, no category. You need clean titles, structured attributes, a category.

- **Volume:** 30 million SKUs per week
- **Task:** narrow, identical output shape every time
- **Latency:** not sensitive, overnight batch
- **Data:** public product info

Answer: Small open-weights model. Managed inference or self-hosted.

Narrow, repetitive task. A tuned small model matches a frontier model here at a fraction of the cost.

Batch means latency is free. Run cheap GPUs at full utilization.

If needed, finetune.

- Use a frontier model once to label more training data, then never again.

Scenario B: Customer Support Chatbot

The problem: Customer-facing bot on app and web. Order status, returns, refunds, delivery issues. Escalates to a human when it cannot resolve.

Volume: 2 million conversations per month, roughly 8 turns each

Task: open-ended conversation in a narrow domain, must call your order and returns APIs

Latency: under 2 seconds, a customer is waiting

Data: order history, addresses, phone numbers.

Accuracy: a wrong refund answer becomes a public tweet

Timeline to ship: 4-6 weeks

Answer: Frontier Proprietary model, accessed through API.

Frontier proprietary model on API, with PII guardrails. Retrieval over your help centre, tool calls into order systems. Do not fine-tune unless absolutely needed.

- Customer phrasing varies far more than the domain suggests. This is not a narrow task.
- Tool-calling reliability is where frontier models are still clearly ahead, and every turn depends on it.
- A refund error costs more than the inference bill.
- PII can be handled through guardrails, not a reason to self-host.
- 16 million turns a month is real money.

Scenario C: Credit Underwriting Assistant

The problem: Regulated lender. Assistant reads borrower documents, bank statements, ITRs, salary slips, and drafts an underwriting summary with flagged risks for the credit officer.

Volume: 500,000 applications per month

Task: extraction and structured summarization over financial documents

Data: full names, PAN, account numbers, income detail

Every output is reviewed by a credit officer before a decision

Answer: Small open-weights model. Self-hosted.

Best available open-weights model that fits your GPU budget, self-hosted in the bank's VPC.

Tracking LLM Progress

Where can we compare LLMs

Two Complementary Ways of Evaluating LLMs

Benchmarks 📊	LMArena (Chatbot Arena)🎭
A set of **standardized leaderboards**, built on trusted datasets (e.g., MMLU, SciCode, Multiloko).	An online platform where users compare model responses in blind, **head-to-head battles**.
Models are tested against fixed tasks → scores are **reproducible and directly comparable**.	Users vote on which response is better → rankings **reflect human preference**.
Strength: Transparent, rigorous, research-grade benchmarking.	**Strength:** Captures real-world qualities like helpfulness, fluency, and reasoning.

Benchmarks tests models in the lab. LMArena tests them in the wild.

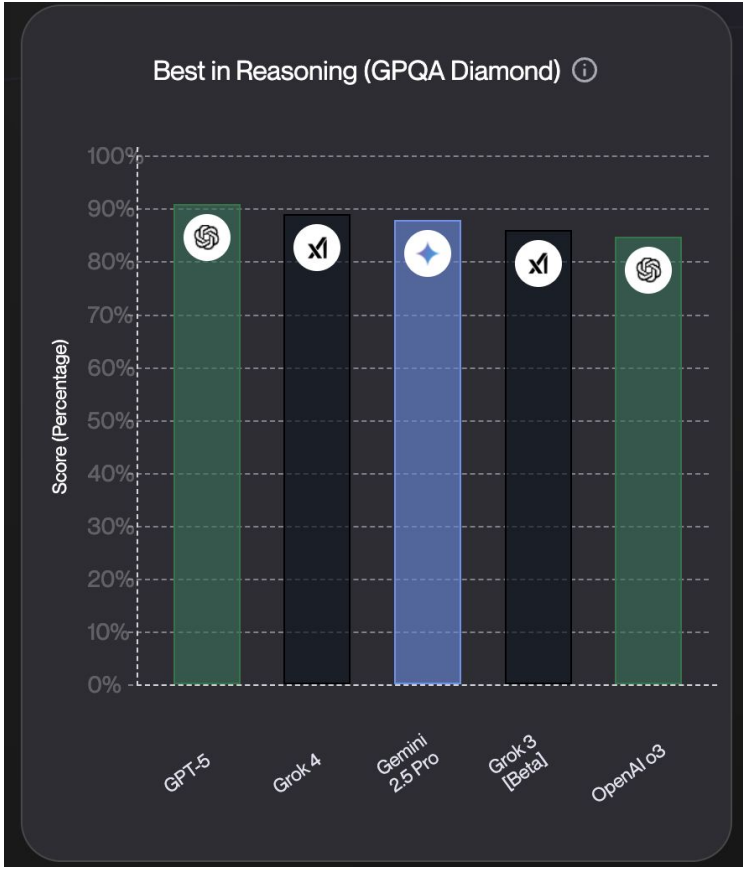
Analyzing model performance - Benchmarks

Standardized tests for evaluating LLMs

Definition: Benchmarks are structured datasets and tasks that act like exam papers for LLMs. They contain fixed inputs and expected outputs, so every model is judged on the same ground.

Purpose: Provide a fair and reproducible way to compare models. Results don't depend on the evaluator. If you run the same model tomorrow, you'll get the same score.

Example Leaderboards - Kaggle | Artificial Analysis | Vellum



Types of benchmarks

Multilingual & Cross-Lingual Capabilities- Test how well models handle multiple languages. e.g., ECLeKTic,MultiLoKo.

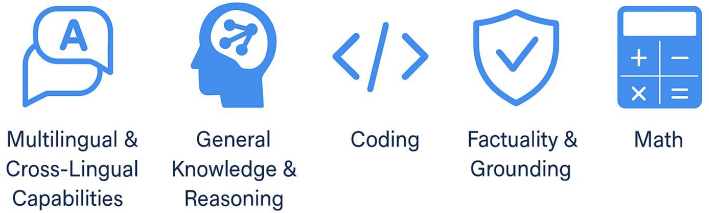
General Knowledge & Reasoning - Assess breadth of knowledge and logical reasoning. e.g., MMLU, GPQA

Coding - Evaluate ability to generate, debug, or explain code. e.g., HumanEval, SciCode

Factuality & Grounding - Check if models give truthful, verifiable answers. e.g., TruthfulQA,FACTS Grounding

Math - Assess arithmetic, problem-solving, and symbolic reasoning. e.g., MathVista, AIME

Different benchmarks test different skills



How are benchmarks designed?

MCQ-style Benchmarks	Natural Language Benchmarks
Predefined question + multiple choices (A/B/C/D)	Open-ended questions, model generates free text
Model must pick the correct option	Answers compared to reference outputs
Easy to grade (accuracy)	Captures fluency, reasoning, creativity.
Limited expressiveness, tests recognition not generation	Harder to evaluate → need Human/LLM judgment
Examples: MMLU, GPQA	Examples: TruthfulQA, HumanEval,

Natural benchmarks require smarter evaluation, unlike MCQs, where accuracy is straightforward.

Community driven LLM Leaderboard – LMArena (Chatbot Arena)

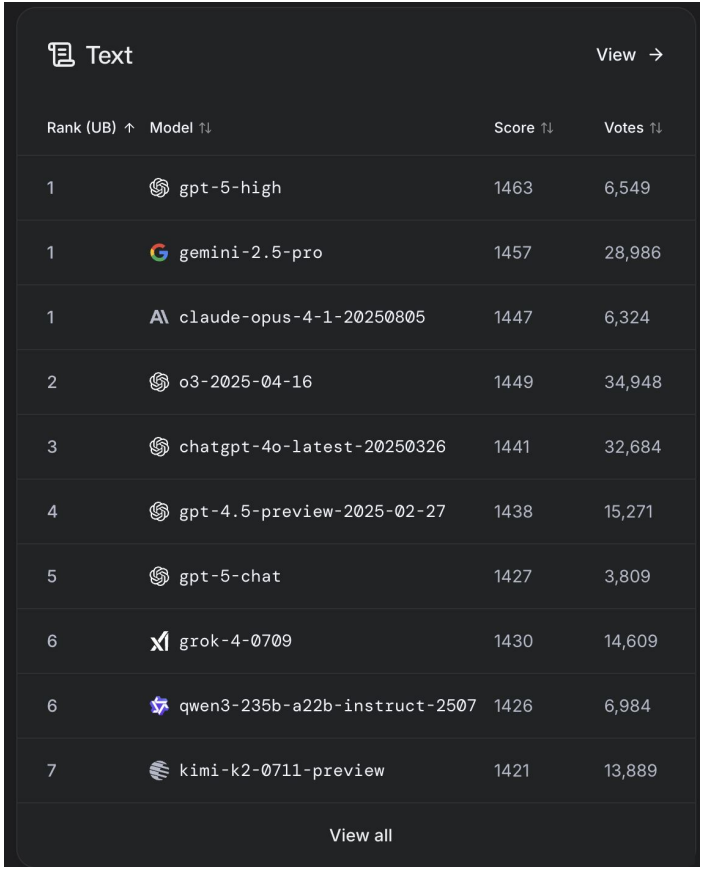
Crowd-sourced evaluation for LLMs

- Head-to-head battles → Two models shown the same prompt, users vote for the better response

- Human preference scores → Rankings based on real user judgments, not just benchmark accuracy

- Dynamic leaderboard → Updates continuously as more comparisons are made

- User-centered performance → Captures qualities like helpfulness, fluency, and reasoning



Questions


thank
you